Eberswalde University for Sustainable Development
Forest Information Technology (FIT)
Applied Programming in Forestry

Project Report

Site-Specific Variability in Height and Diameter Growth of *Melia volkensii* Woodlots in Kilifi County, Kenya.

Submitted by:
Julie Chango Litu

Submission date:
20/02/2025

Table of Contents

1. Introduction 2
1.1 Research question and hypothesis 2
2. Materials and Methods 4
2.1 Study Area Description 4
2.2 Data Analysis 4
2.2.1 Data preprocessing 4
2.2.2 Data Visualization and descriptive statistics 4
2.2.3 Statistical methods 5
2.3 Input and output data 5
3. Results 6
3.1 Tree DBH BoxPlots 6
3.2 Tree Height BoxPlots 6
3.3 Statistical tests 6
4. Discussion and Conclusion 7
Appendix 13

List of Tables

Table 1: Descriptive statistics summary for DBH *8*
Table 2: Descriptive statistics summary for tree height *9*
Table 3: ANOVA results for DBH *10*
Table 4: Tukey HSD results for DBH *10*
Table 5: ANOVA results for tree height *10*
Table 6: Tukey HSD results for tree height *10*

List of Figures

Figure 1: Box plot of DBH by site *8*
Figure 2: Box plot of tree height by site *9*

1. Introduction

Melia volkensii is a highly adapted tree species endemic to the Kenyan drylands given its drought-tolerant attributes. It has received a high preference for cultivation on farms by smallholder farmers due to its fast growth, short rotation period of about 10 to 15 years, and highly profitable hardwood timber which is resistant to termites and decay (Luvanda et al., 2015; Wekesa et al., 2012). Additionally, *Melia volkensii* plays an important role in providing other ecosystem services including fodder and medicine provision, carbon sequestration, and soil improvement (Mwadalu et al., 2021a; Ndufa et al., 2017). Despite the widespread use and cultivation of *Melia volkensii* by smallholder farmers, its growth performance can vary significantly across different sites, influenced by factors such as climate, soil properties, management practices, and competition for resources.

Understanding these variations is vital for effective forest management, and productivity maximization**.** By analysing tree growth differences across sites using key metrics such as Diameter at Breast Height (DBH) and height (Skovsgaard & Vanclay, 2008), researchers and forest managers can understand how different site conditions affect tree development and make informed decisions about site selection, species suitability, and sustainable management practices.

Numerous research has been done on *Melia Volkensii* in Kenya by Kenya Forestry Research Institute (Ochieng et al., 2018) exploring several different topics given its importance in the Kenyan drylands. However, no study has focused on the impact of site conditions on *Melia volkensii* tree growth performance. This study seeks to address this knowledge gap by investigating the growth variations of 4-year-old *Melia volkensii* woodlots in three different sites in Kilifi County, Kenya by examining DBH and Height across locations. Statistical analyses, including ANOVA and Tukey's HSD test, will be used to determine whether site-specific differences in tree growth are statistically significant. The findings will contribute to improved site selection strategies, growth optimization, and sustainable management practices of *Melia volkensii* in Kilifi County, Kenya.

1.1 Research question and hypothesis

This study aims to answer the following questions:

1. How do DBH and height of *Melia volkensii* trees vary across different sites in Kilifi County?
2. Are there statistically significant differences in DBH and height between the different sites?

To test these questions, the study proposes the following hypotheses:

- **H_0:** There is no significant difference in DBH and height among *Melia volkensii* trees across different sites.

- **H_1:** There is a significant difference in DBH and height among *Melia volkensii* trees across different sites.

2. Materials and Methods

2.1 Study Area Description

Kilifi County is in the coastal region of Kenya bordering the Indian Ocean to the east. It covers an area of 12,245.90 $km^2$and lies between longitude 39°05" and 40°14" east and between latitude 2°20" and 4°0" south. The county land gradually rises from sea level to 900m above sea level from east to west. The annual precipitation of the county ranges between 300mm - 1300 mm with an annual mean temperature of 27 °C. A bimodal rainfall regime is experienced in the county with long rains from March to May and short rains between October and December (Mwadalu et al., 2021).

Kilifi County was selected as the study area because of the following reasons. First, the establishment of *M. volkensii* woodlot plantations have been widely promoted by private sector organizations such as Komaza Forestry Ltd since the year 2004. Second, more smallholder farmers have ventured into *M. volkensii* farming (Wanjira & Muriuki, 2020).

2.2 Data Analysis

This project analysed tree height and DBH data collected from one-acre, four-year-old *Melia volkensii* woodlots across three different sites in Kilifi County. Site 1 comprised a sample size of 21 trees, while Site 2 and 3 comprised of a sample size of 20 trees each. The data was stored in a CSV file format.

The data analysis was done using python. The analysis entailed: data preprocessing, data visualization and statistical tests using different python libraries to determine the variability in tree height and diameter growth across the three sites. The python libraries imported for the analysis included: pandas, matplotlib, seaborn, numpy, scipy, and statsmodels.

2.2.1 Data preprocessing

The Kilifi_data csv file containing *Melia volkensii* dataset was imported, read, and processed using Pandas. The 'Site' column was converted into a categorical data type using Panda's "astype()" function for correct statistical analysis. The missing values were then checked using Pandas "isna()" function and counted for each column using the "sum()" function. These steps are important in the analysis because incorrect data types and missing values may create problems in the codes when performing statistical tests.

2.2.2 Data Visualization and descriptive statistics

The Boxplots were created using Seaborn and Matplotlib python libraries to visualize the distribution of tree diameter and height across the three sites. The matplotlib.pyplot was used to create the

interactive visualization and Seaborn's "boxplot()" function was used to create the boxplots. Descriptive statistics for both DBH and tree height were computed using pandas and numpy python libraries. The "describe()" function was used.

2.2.3 Statistical methods

Three statistical tests were applied for this study. The p-value (α = 0.05) (Kim, 2017) was used for all the statistical tests.

Levene's test

Levene's test was conducted using scipy.stats python library. The test was separately conducted for DBH and Height to test for the homogeneity of variances across the three sites and the "stats.levene()" function was used for the analysis. The results were printed using the "print()" function.

One-way ANOVA

A one-way ANOVA was conducted separately for DBH and Height to determine if there were statistically significant differences in mean values among the three sites. The statsmodels python library was used. The "smf.ols()" function was applied to fit the ANOVA model and the "sm.stats.anova_lm()" function was applied for the ANOVA table computation. "The fit()" function was applied to estimate the model parameters.

The ANOVA model(Kim, 2017) was structured as:

$$y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$$

Where y_{ij} is DBH or Height, α_i is the site, μ is the overall population mean, and $\varepsilon_{ij}$is the residuals

Tukey's Honest Significant Difference (HSD) Test

The Tukey's HSD test was applied as a post-hoc analysis to identify which pairs of sites had statistically different means in cases where ANOVA indicated significant differences. The "pairwise_tukeyhsd()" function was applied to perform pairwise comparisons to determine the specific sites with significantly different DBH and tree height values.

2.3 Input and output data

The imported csv file was converted into a Pandas DataFrame. The input data for the codes included the DBH and height with numeric (float) data type, and site column with categorical data type. The output data included boxplot graphs, p-values and F- statistic values. Both the p-values and F- statistic values are of numeric (float) data type.

3. Results

3.1 Tree DBH BoxPlots

There is a noticeable difference in the median DBH across sites 1, 2 and 3(Figure 1). The median DBH is highest in Site 3 with a value of 13.95, followed by Site 1 with a value of 12.30, and lowest in Site 2 with a value of 10.65(Table 1). Site 2 exhibits a larger spread in DBH compared to Site 1 and Site 3. Outliers are also observed in sites 1 and 3.

3.2 Tree Height BoxPlots

There is a noticeable difference in the median tree height across sites 1, 2 and 3(Figure 2). The median height is highest in Site 3 with a value of 6.30, followed by Site 1 with a value of 5.70, and lowest in Site 2 with a value of 5.15(Table 2). Site 2 exhibits a larger spread in height compared to Site 1 and Site 3. Outliers are only observed in sites 1(Figure 2).

The patterns exhibited in the boxplots suggest variation in both tree DBH and tree height across the three sites, which is further tested using ANOVA and Tukey's post hoc analysis.

3.3 Statistical tests

One-way ANOVA and a post-hoc Tukey's HSD tests were conducted separately for DBH and Height to determine if there were statistically significant differences in mean values among the three sites. The p-value ($\alpha = 0.05$) helps determine whether we reject or fail to reject the null hypothesis (H_0)(Foster, 2001). If $p \leq 0.05$ we reject the H_0 and if $p > 0.05$ we fail to reject $\mathbf{H_0}$**.**

P-values of 0.000582 and 0.000998 (Table 3 and Table5) which are below the significance threshold ($\alpha = 0.05$) were obtained for DBH and tree height respectively, indicating significant differences in the DBH, and tree height means across the tree sites. In this case, the null hypothesis was rejected.

The Tukey's HSD test confirmed a significant difference between Sites 1 & 2 with a P-value of 0.0377, and Sites 2 & 3 with a P-value of 0.0007 for the height. No significant difference was found for height between sites 1 & 3 for DBH with a P-value of 0.3363 (Table 6). A significant difference was only found between Sites 2 & 3 with a P-value of 0.0004 for the DBH and no significant differences were found between Sites 1 & 2 with a P-value of 0.0928, and Sites 1 & 3 with a P-value of 0.1113 for the DBH (Table 4).

4. Discussion and Conclusion

The study results indicate a significant variation of *M. volkensii* tree DBH and height across the three different sites. The boxplots visually suggest these differences, which were statistically confirmed by ANOVA and post hoc Tukey's HSD tests. The statistical analysis confirmed that *M. Volkensii* trees in site 3 had significantly larger and taller trees exhibiting better growth compared to sites 1 and 2. Site 2 had significantly smaller trees exhibiting poor growth, while site 1 exhibited moderate growth.

This could be attributed to various factors including environmental conditions, soil conditions and management practices (Skovsgaard & Vanclay, 2008). These results underscore the importance of considering site conditions for effective management and growth optimization of *M. Volkensii* woodlots in Kilifi County. Future research could additionally explore the specific factors influencing *M. volkensii* tree growth on the different sites to further validate these findings.

Table 1: Descriptive statistics summary for DBH

	count	**mean**	**std**	**min**	**25%**	**50%**	**75%**	**max**
Site								
1	21.0	11.761905	2.481628	6.8	10.900	12.30	13.000	15.7
2	20.0	10.205000	2.468000	6.1	8.475	10.65	11.475	15.0
3	20.0	13.255000	2.036115	7.8	12.050	13.95	14.500	16.6

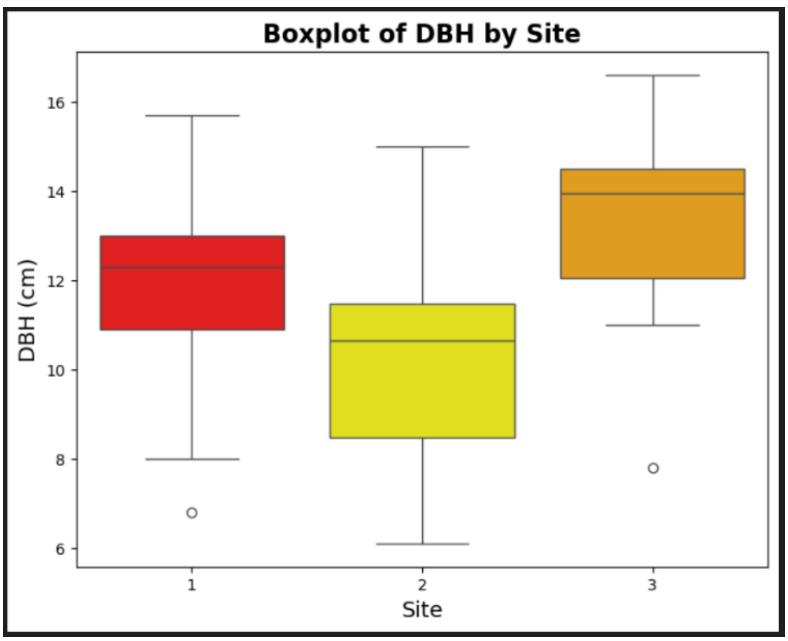

Figure 1: Box plot of DBH by site

Table 2: Descriptive statistics summary for tree height

	count	**mean**	**std**	**min**	**25%**	**50%**	**75%**	**max**
Site								
1	21.0	5.804762	0.834551	4.0	5.4	5.70	6.2	7.6
2	20.0	5.130000	0.957024	3.6	4.3	5.15	5.7	7.0
3	20.0	6.185000	0.765902	4.8	5.9	6.30	6.7	7.6

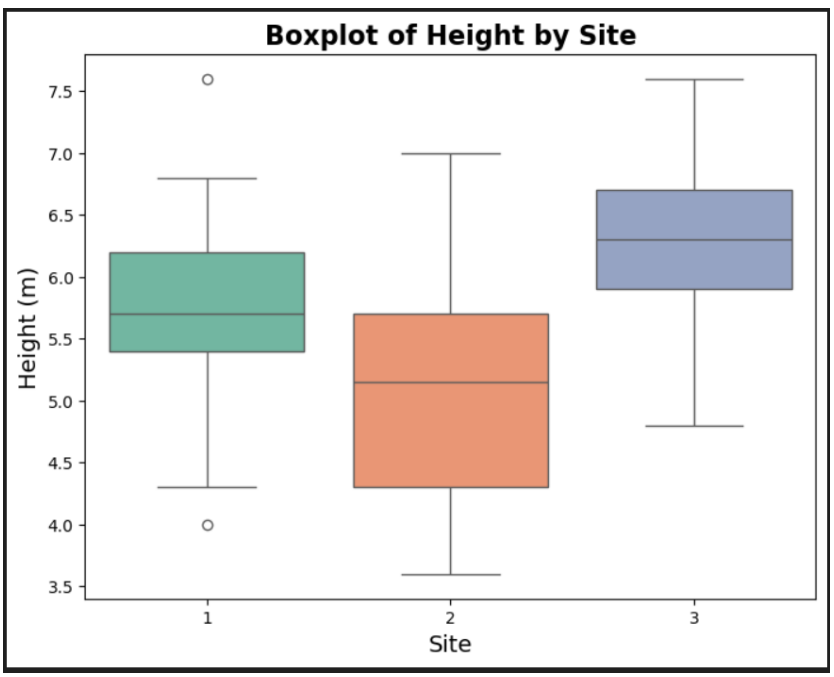

Figure 2: Box plot of tree height by site

Table 3: ANOVA results for DBH

```
ANOVA results for DBH:
               sum_sq    df         F    PR(>F)
Site       93.039017   2.0  8.493544  0.000582
Residual  317.668524  58.0       NaN       NaN
```

Table 4: Tukey HSD results for DBH

```
Tukey HSD results for DBH:
 Multiple Comparison of Means - Tukey HSD, FWER=0.05
====================================================
group1 group2 meandiff p-adj   lower  upper  reject
----------------------------------------------------
     1      2  -1.5569 0.0928 -3.3157 0.2019  False
     1      3   1.4931 0.1113 -0.2657 3.2519  False
     2      3     3.05 0.0004  1.2699 4.8301   True
----------------------------------------------------
```

Table 5: ANOVA results for tree height

```
ANOVA results for Height:
               sum_sq    df         F    PR(>F)
Site       11.428878   2.0  7.802747  0.000998
Residual   42.477024  58.0       NaN       NaN
```

Table 6: Tukey HSD results for tree height

```
Tukey HSD results for Height:
 Multiple Comparison of Means - Tukey HSD, FWER=0.05
====================================================
group1 group2 meandiff p-adj   lower   upper  reject
----------------------------------------------------
     1      2  -0.6748 0.0377 -1.3179 -0.0316   True
     1      3   0.3802 0.3363 -0.2629  1.0234  False
     2      3    1.055 0.0007  0.4041  1.7059   True
----------------------------------------------------
```

References

Foster, J. R. (2001). Statistical power in forest monitoring. *Forest Ecology and Management*, *151*(1–3), 211–222.

Kim, T. K. (2017). Understanding one-way ANOVA using conceptual figures. *Korean Journal of Anesthesiology*, *70*(1), 22–26. https://doi.org/10.4097/kjae.2017.70.1.22

Luvanda, A. M., Musyoki, J. K., Cheboiwo, J. K., Wekesa, L., & Ozawa, M. (2015). *An assessment of the socio-economic importance of Melia volkensii based enterprises in Kenya*.

Mwadalu, R., Gathara, M., Muturi, G., & Mbuvi, M. T. E. (2021a). *POTENTIAL OF CASUARINA EQUISETIFOLIA AND MELIA VOLKENSII TREE SPECIES IN IMPROVING SOIL FERTILITY IN KWALE AND KILIFI COUNTIES, KENYA*.

Ndufa, J. K., Miyashita, H., Kigwa, B., Auka, S., & Muturi, G. (2017). XI Estimating biomass and carbon stocks of Melia volkensii plantation in dryland landscapes of Kenya. *Proceedings of International Conference on the "Project on Development of Drought Tolerant Trees for Adaptation to Climate Change in Drylands of Kenya" 13-16 February 2017 at KEFRI Headqurters Muguga*, 106.

Ochieng, D., Kamondo, B., & Muturi, G. M. (2018). *Proceedings of International Conference on 'Project on Development of Drought Tolerant Trees for Adaptation to Climate Change in Drylands of Kenya.'* International Conference on 'Project on Development of Drought Tolerant Trees for Adaptation to Climate Change in Drylands of Kenya,' Kenya.

Skovsgaard, J. P., & Vanclay, J. K. (2008). Forest site productivity: A review of the evolution of dendrometric concepts for even-aged stands. *Forestry*, *81*(1), 13–31. https://doi.org/10.1093/forestry/cpm041

Wanjira, E. O., & Muriuki, J. (2020). *Review of the Status of AgroforestryPractices in Kenya. Background study for preparation of Kenya National Agroforestry Strategy (2020 – 2030)*. *March 2020*, 1–47. https://doi.org/10.13140/RG.2.2.35739.59681

Wekesa, L., Muturi, G., Mulatya, J., Esilaba, A. O., Keya, G. A., & Ihure, S. (2012). *Economic viability of Melia volkensii (Gurkii) production on smallholdings in drylands of Kenya.*

Appendix

4 year old *M.volkensii* Site 1

4 year old *M.volkensii* Site 2